

Chapter 7

Data Communications and Networking

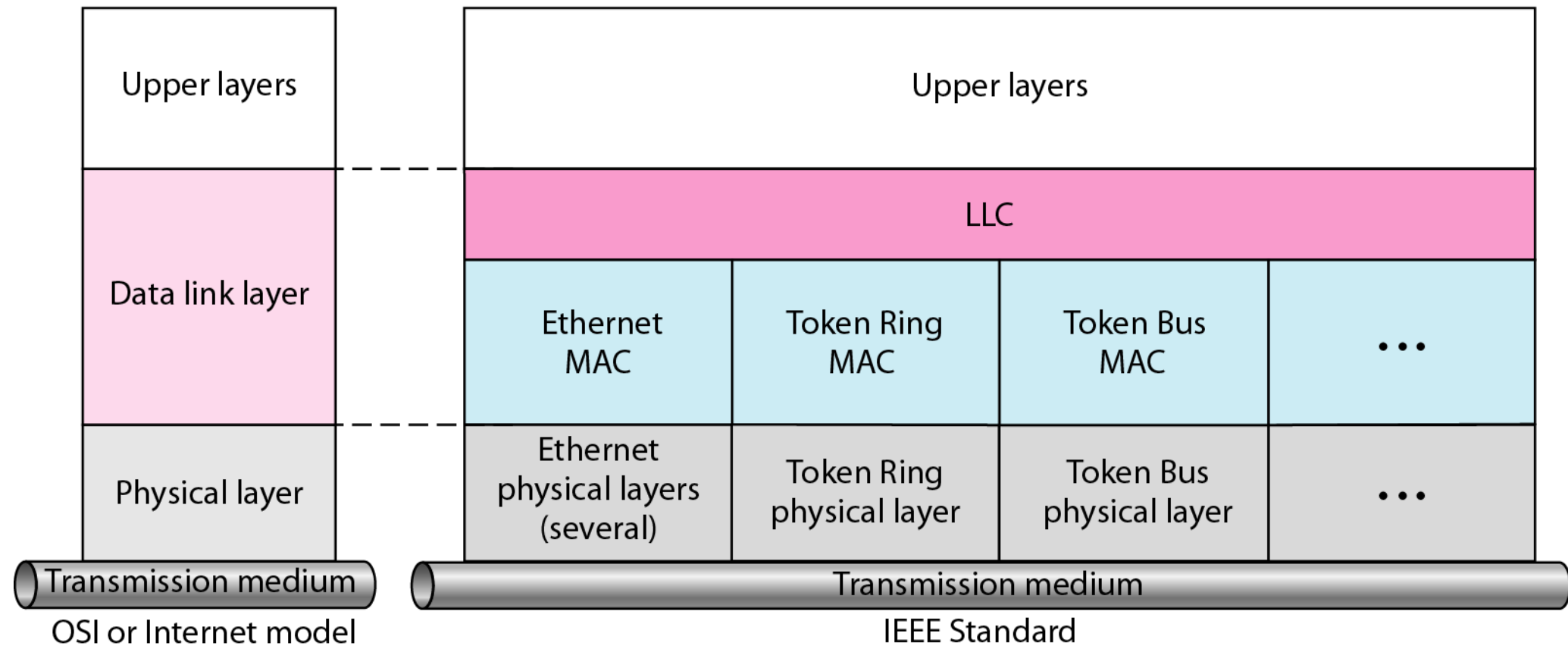
Standardized Networks

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers.
- Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.

IEEE standard for LANs

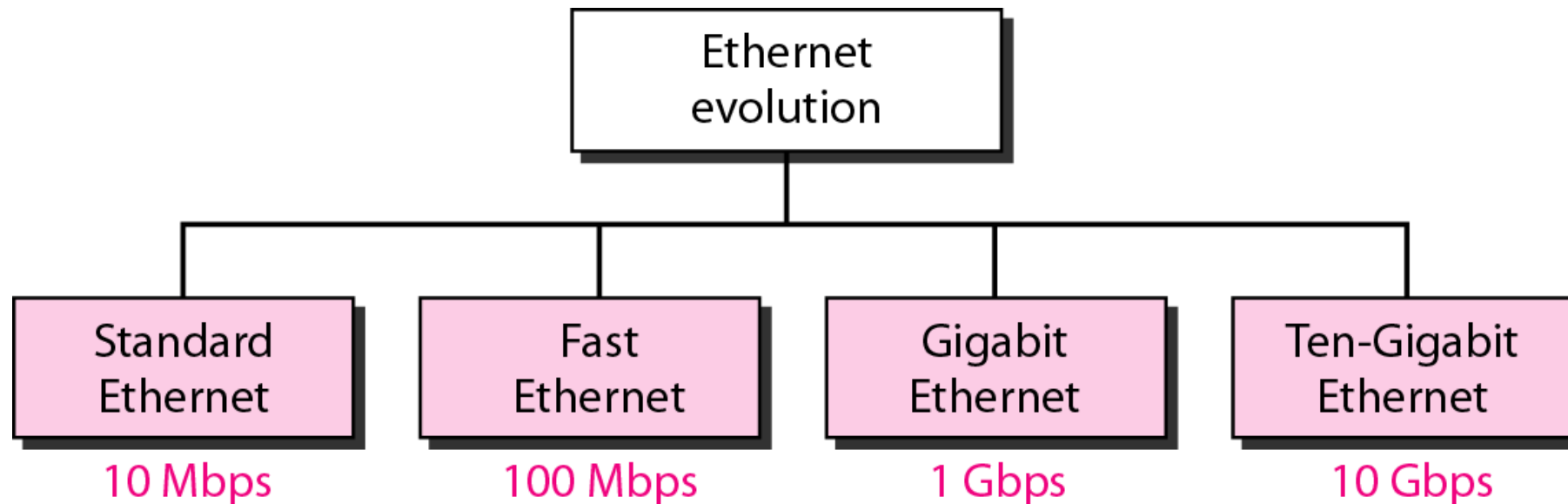
LLC: Logical link control

MAC: Media access control





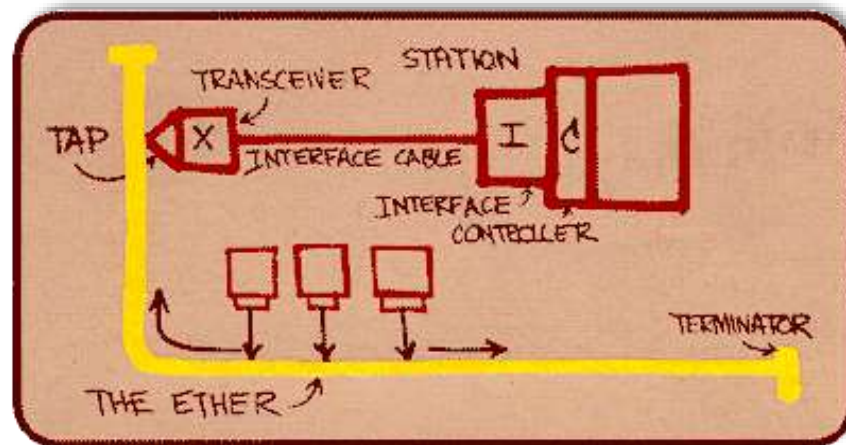
- The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through several generations.
- We briefly discuss the Standard Ethernet in this section.



“dominant” wired LAN technology:

- first widely used LAN technology
- simpler, cheap
- kept up with speed race: 10 Mbps – 400 Gbps
- single chip, multiple speeds (e.g., Broadcom BCM5761)

Metcalfe's Ethernet sketch



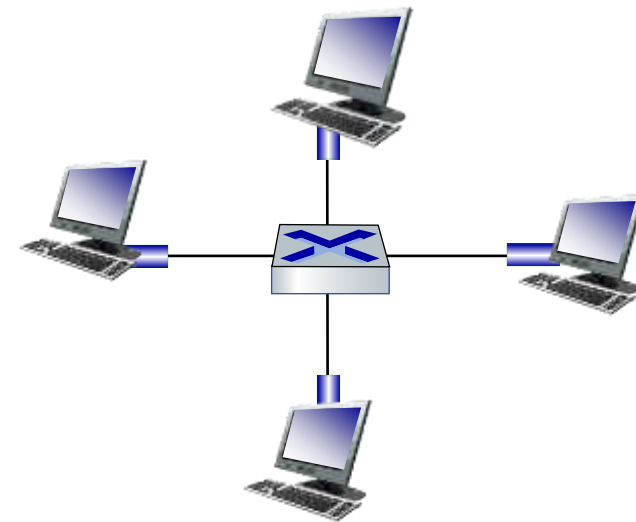
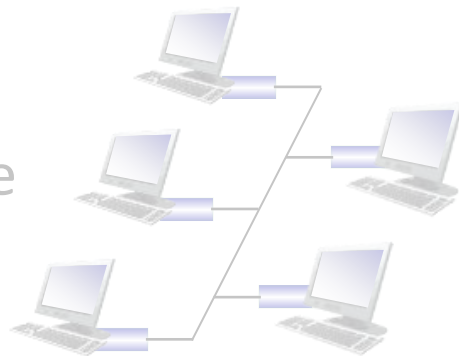
Bob Metcalfe: Ethernet co-inventor,
2022 ACM Turing Award recipient



Ethernet: physical topology

- **bus:** popular through mid 90s
 - all nodes in same collision domain (can collide with each other)
- **switched:** prevails today
 - active link-layer 2 *switch* in center
 - each “spoke” runs a (separate) Ethernet protocol (nodes do not collide with each other)

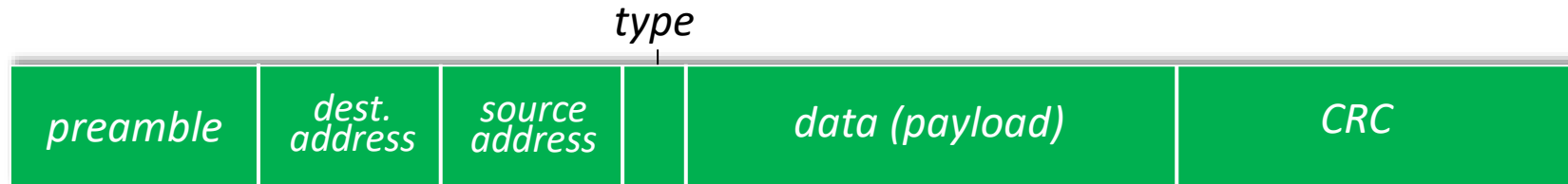
bus: coaxial cable



switched

Ethernet frame structure

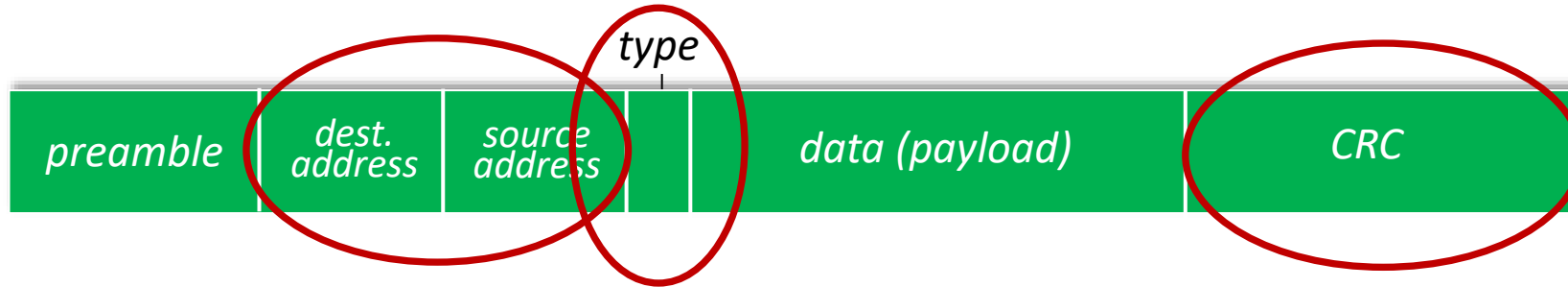
sending interface encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



preamble:

- used to synchronize receiver, sender clock rates
- 7 bytes of 10101010 followed by one byte of 10101011

Ethernet frame structure (more)



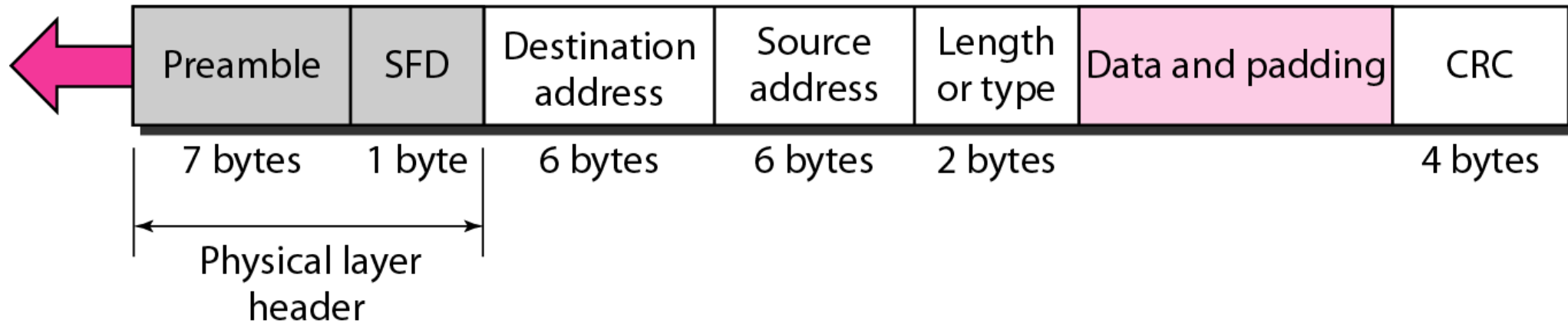
- **addresses:** 6 byte source, destination MAC addresses
 - if adapter receives frame with matching destination address, or with broadcast address (e.g., ARP packet), it passes data in frame to network layer protocol
 - otherwise, adapter discards frame
- **type:** indicates higher layer protocol
 - mostly IP but others possible, e.g., Novell IPX, AppleTalk
 - used to demultiplex up at receiver
- **CRC:** cyclic redundancy check at receiver
 - error detected: frame is dropped

Ethernet (IEEE 802.3)

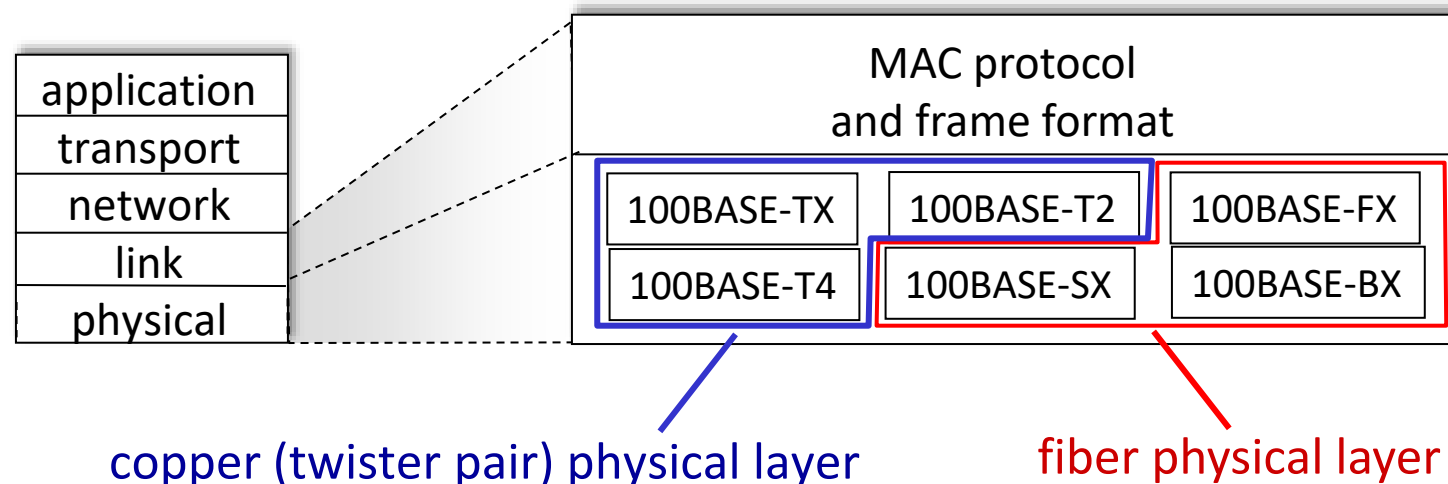
- 802.3 MAC frame

Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)

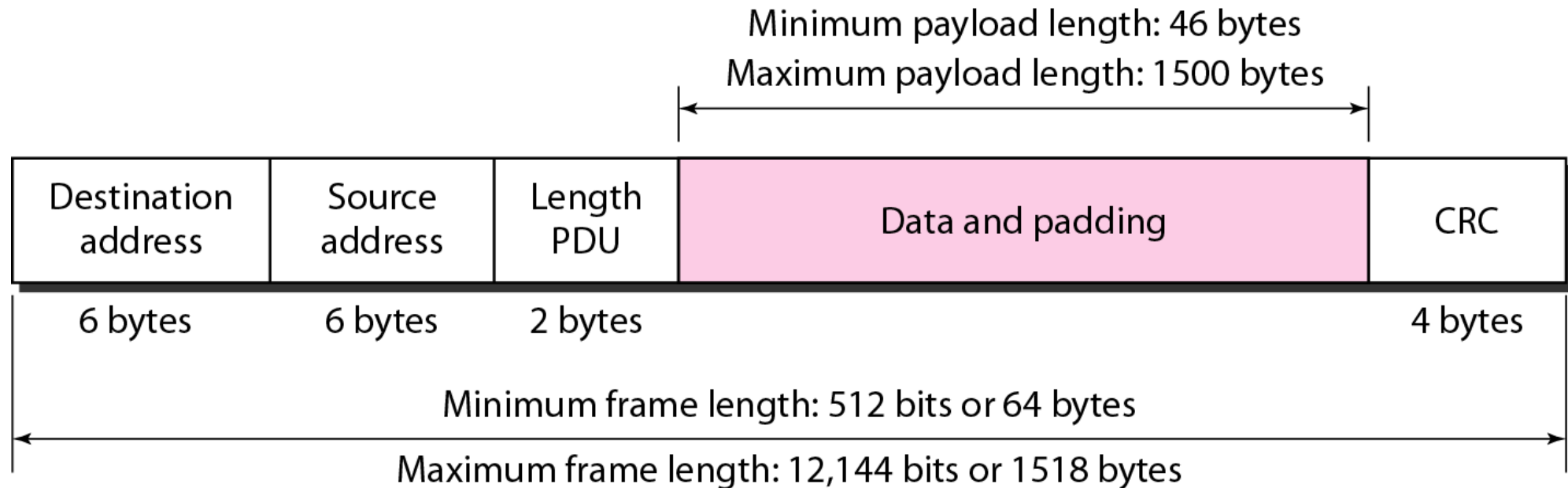


- *many* different Ethernet standards
 - common MAC protocol and frame format
 - different speeds: 2 Mbps, ... 100 Mbps, 1Gbps, 10 Gbps, 40 Gbps, 80 Gbps
 - different physical layer media: fiber, cable

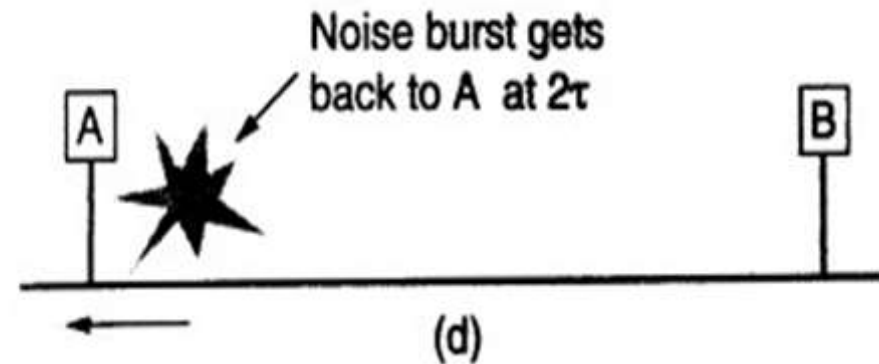
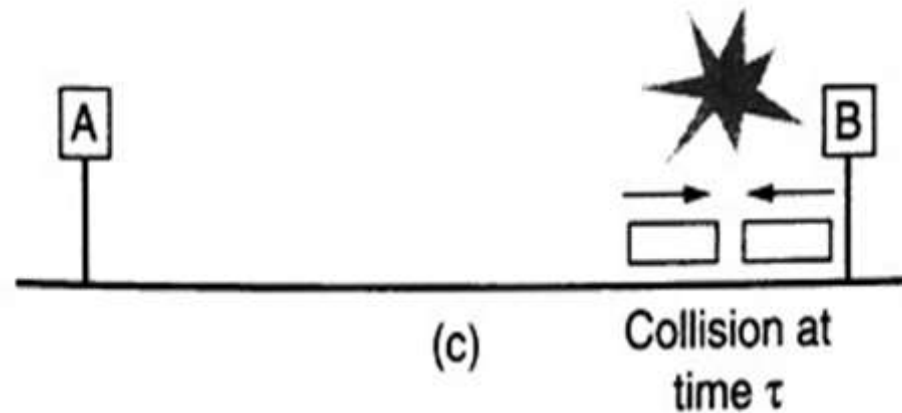
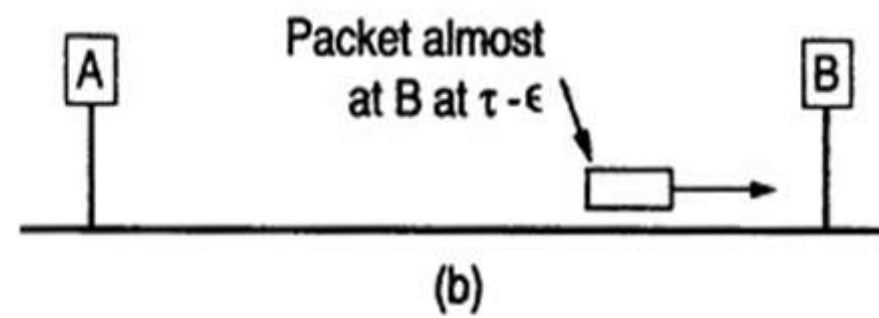
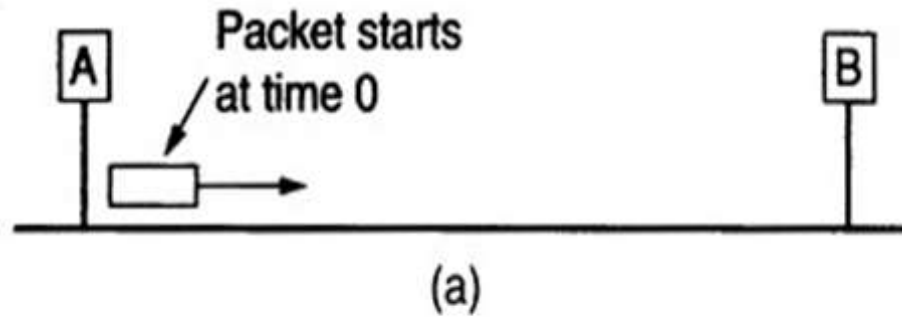


Ethernet (IEEE 802.3)

- Minimum and maximum lengths



Determination of the Minimum Frame Size



Collision detection can take as long as 2τ .

- How to determine the minimum frame size?

Example



- Cable length: 1km
 - Network speed: 100 MB/s
 - Signal speed in the cable: $\frac{2}{3} \times \text{light speed}$ (3×10^8 m/s)
 - What is the minimum frame size?
-
- Propagation delay (round trip): $2 \times 1 / (2 \times 10^5) = 1 \times 10^{-5} s$
 - Number of bits: $100 \times 10^6 \times 1 \times 10^{-5} = 1000$ bits = 125 bytes
 - The minimum frame size is 125 bytes.



- Example of an Ethernet address in hexadecimal notation

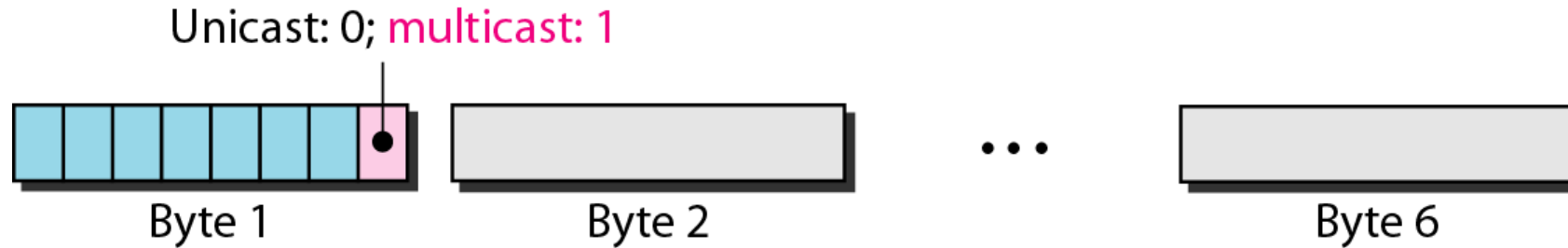
06 : 01 : 02 : 01 : 2C : 4B



6 bytes = 12 hex digits = 48 bits

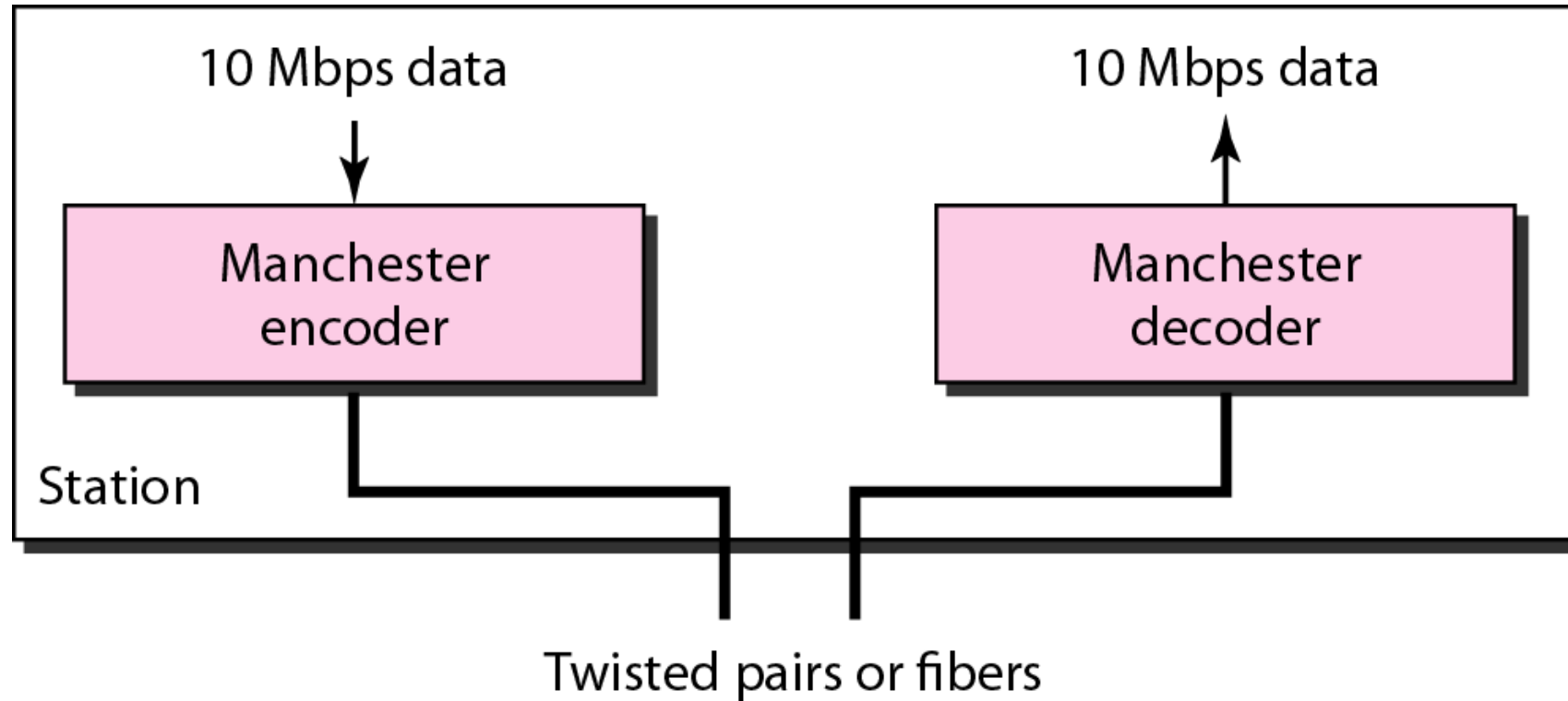
Ethernet (IEEE 802.3)

- Unicast and multicast addresses

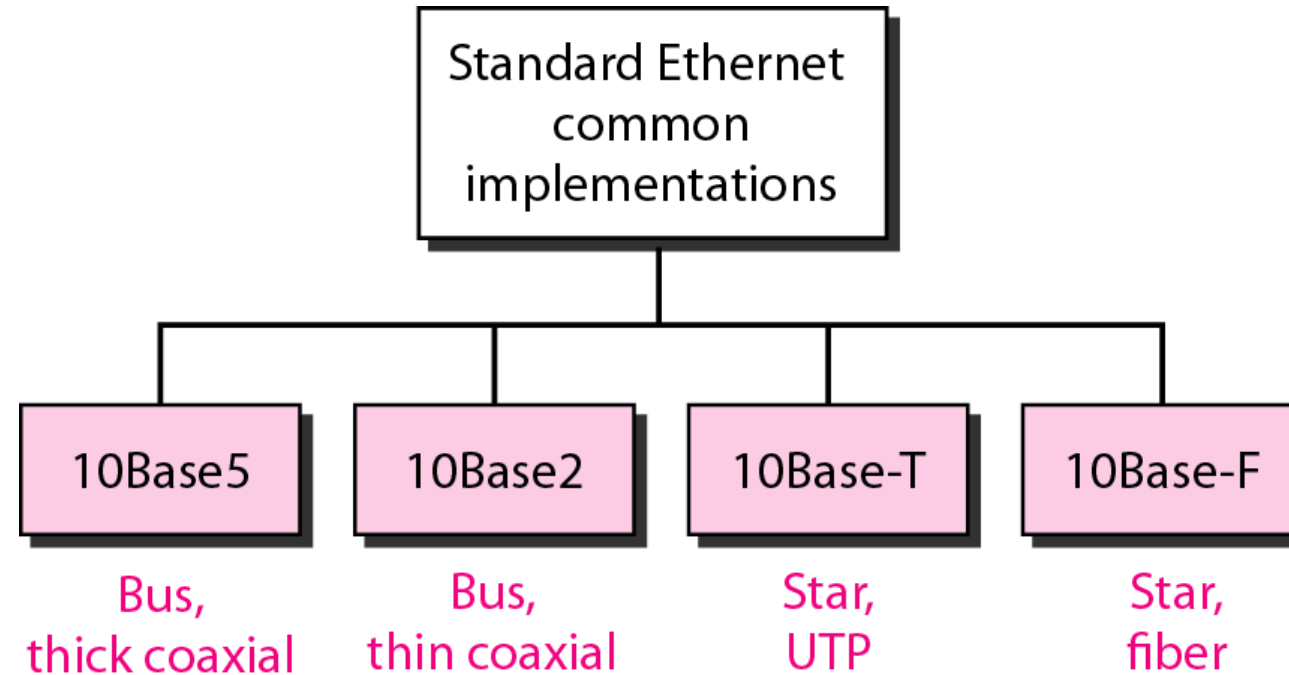


- The least significant bit of the first byte defines the type of address. If the bit is 0, the address is unicast; otherwise, it is multicast.
- The broadcast destination address is a special case of the multicast address in which all bits are 1s.

Encoding in a Standard Ethernet Implementation



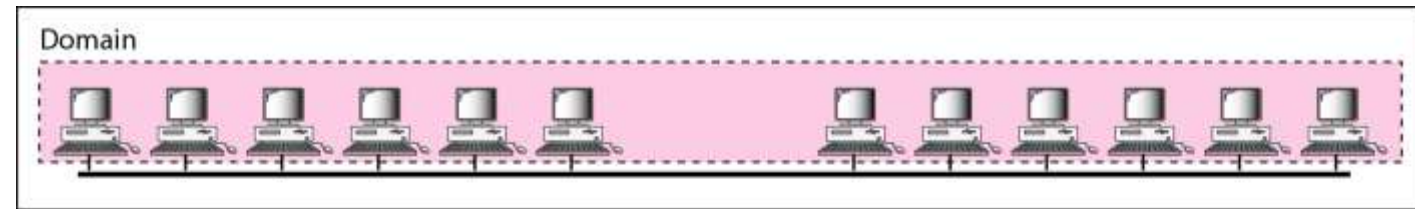
Categories of Standard Ethernet



<i>Characteristics</i>	<i>10Base5</i>	<i>10Base2</i>	<i>10Base-T</i>	<i>10Base-F</i>
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester

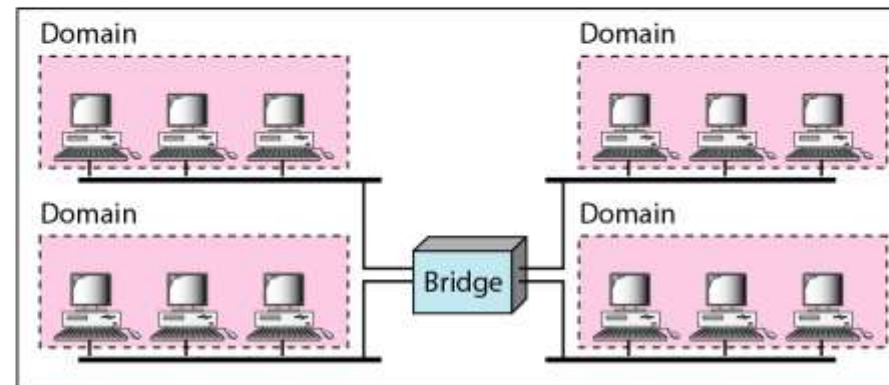
Changes in The Standard

- The 10-Mbps Standard Ethernet has gone through several changes before moving to the higher data rates.
- These changes actually opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs.



a. Without bridging

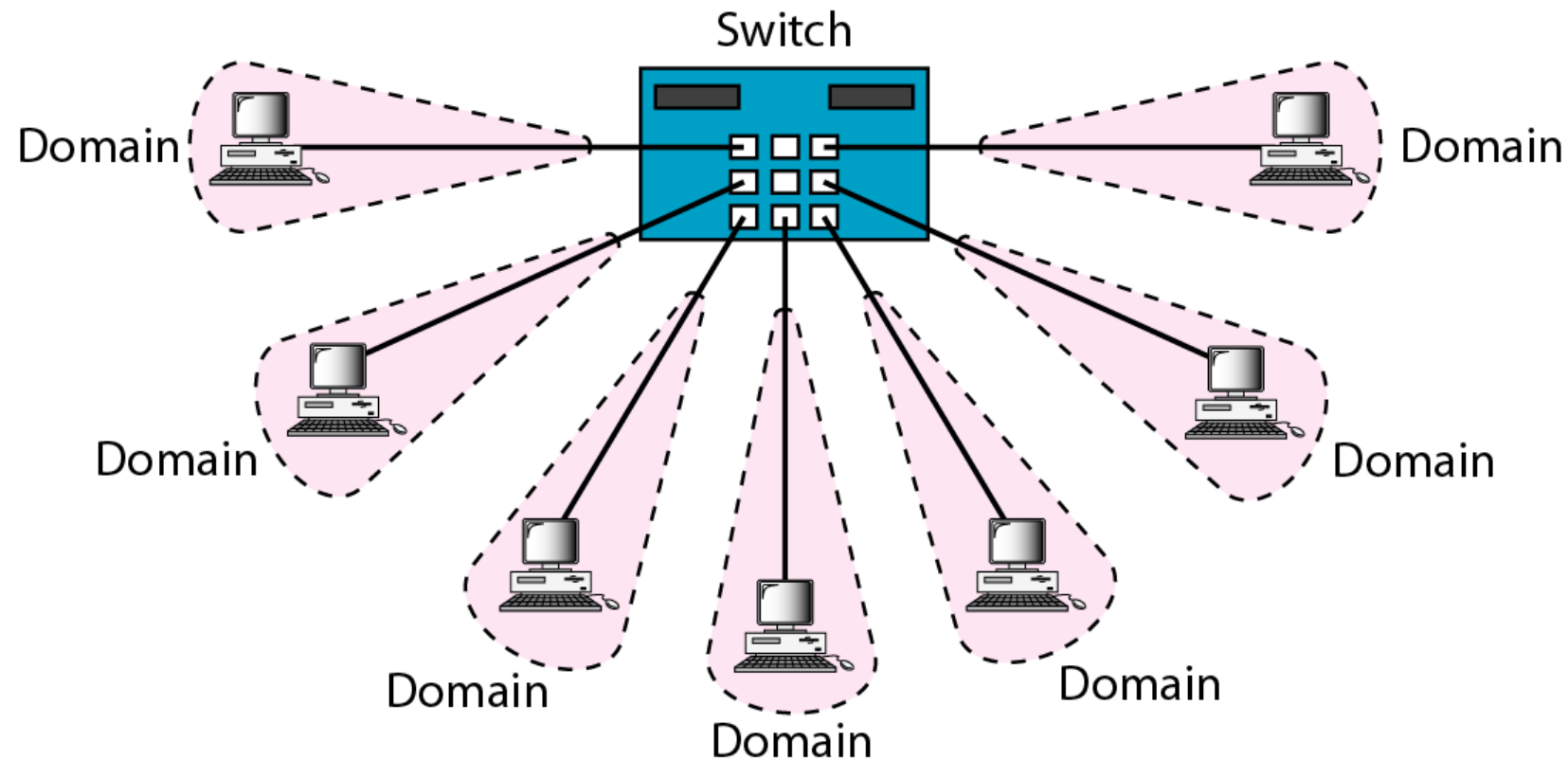
- Bridged network



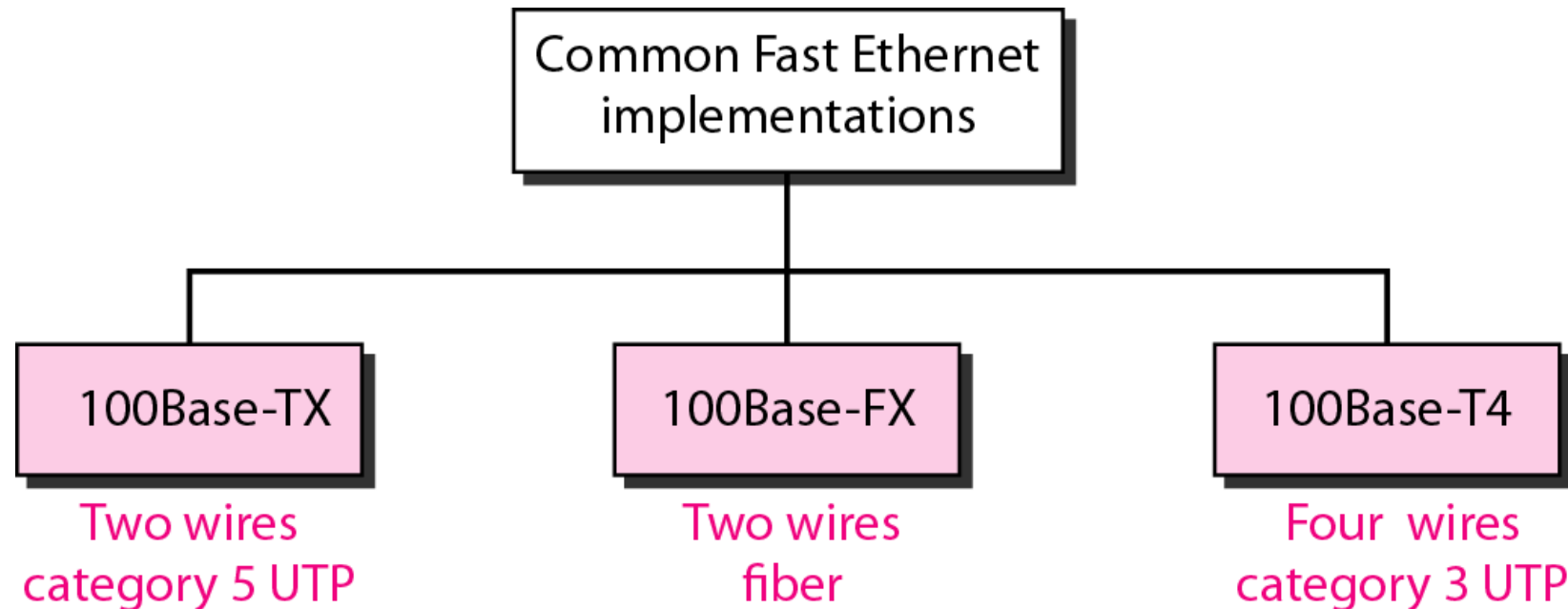
b. With bridging

Changes in The Standard

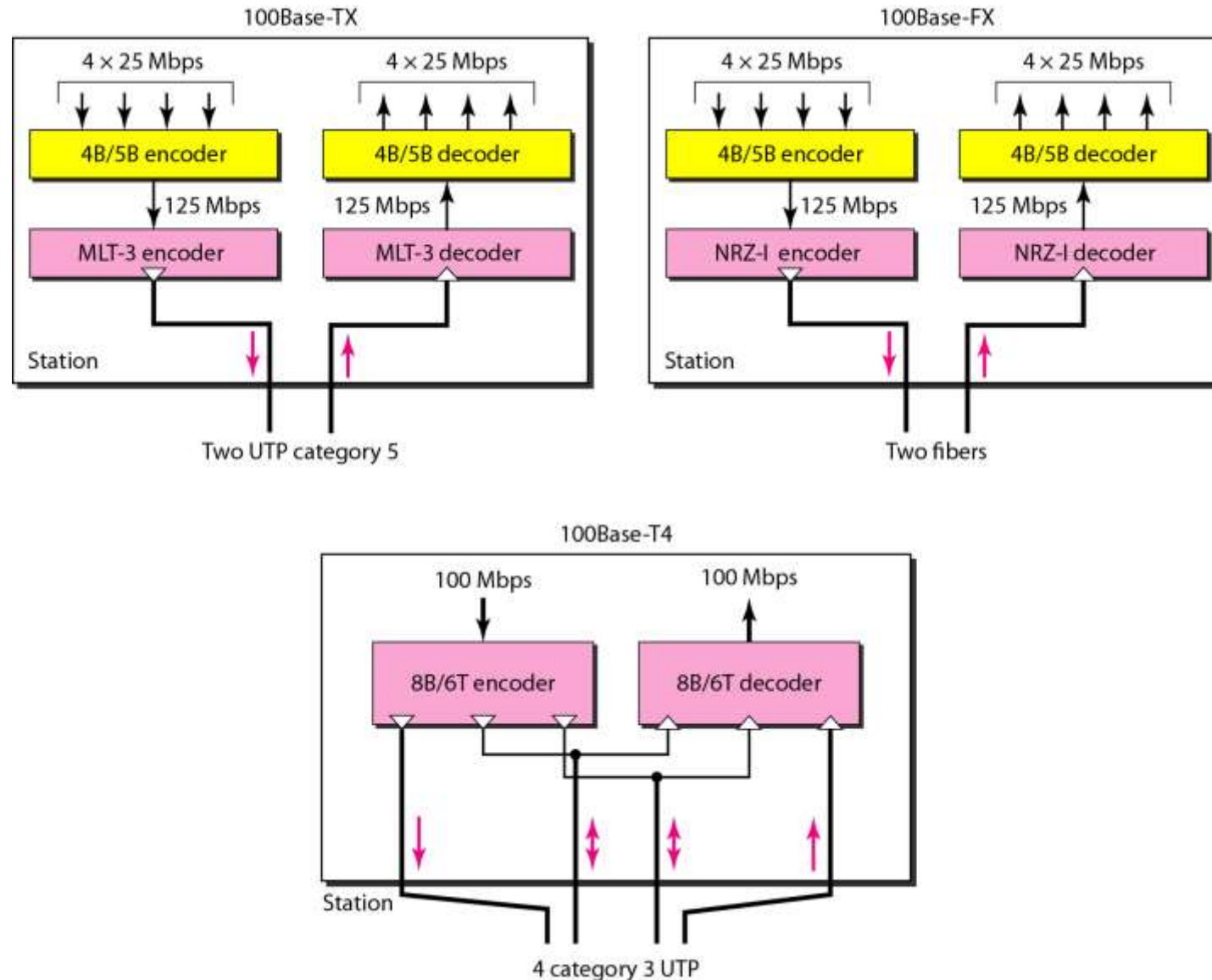
- Switched Ethernet



- Fast Ethernet was designed to compete with LAN protocols such as FDDI or Fiber Channel.
- IEEE created Fast Ethernet under the name 802.3u.
- Fast Ethernet is backward-compatible with Standard Ethernet, but it can transmit data 10 times faster at a rate of 100 Mbps.



Encoding for Fast Ethernet implementation

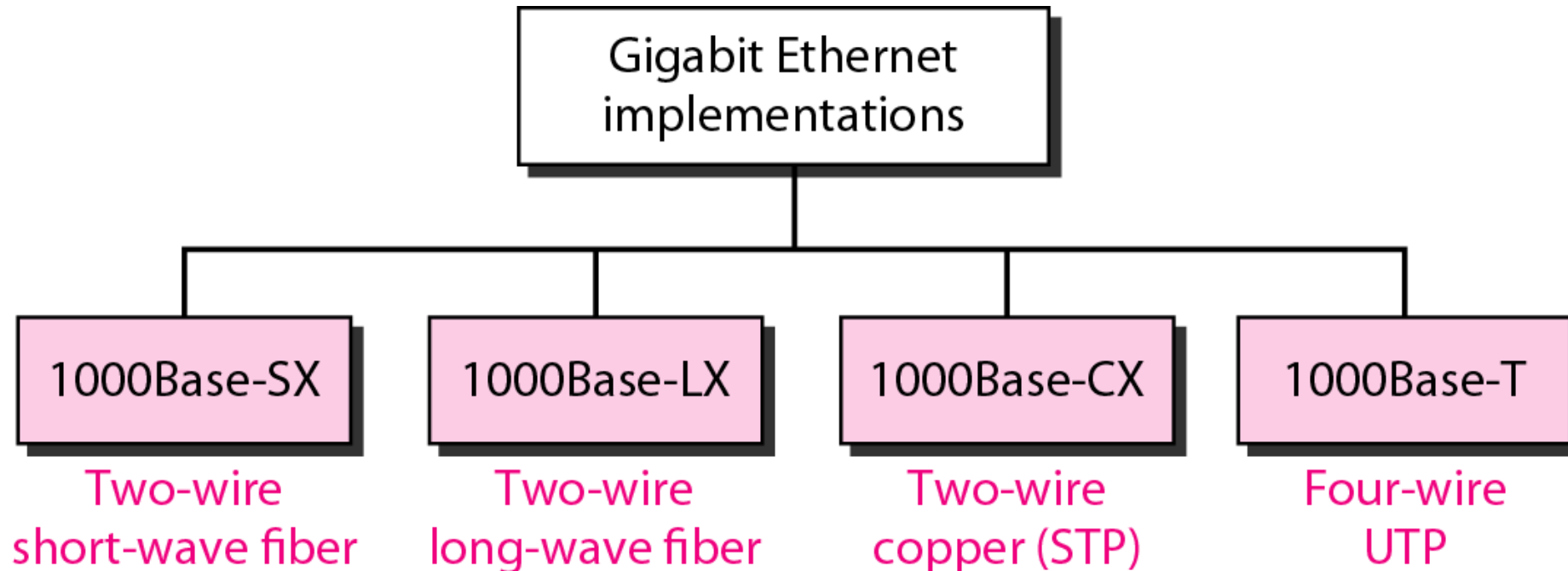




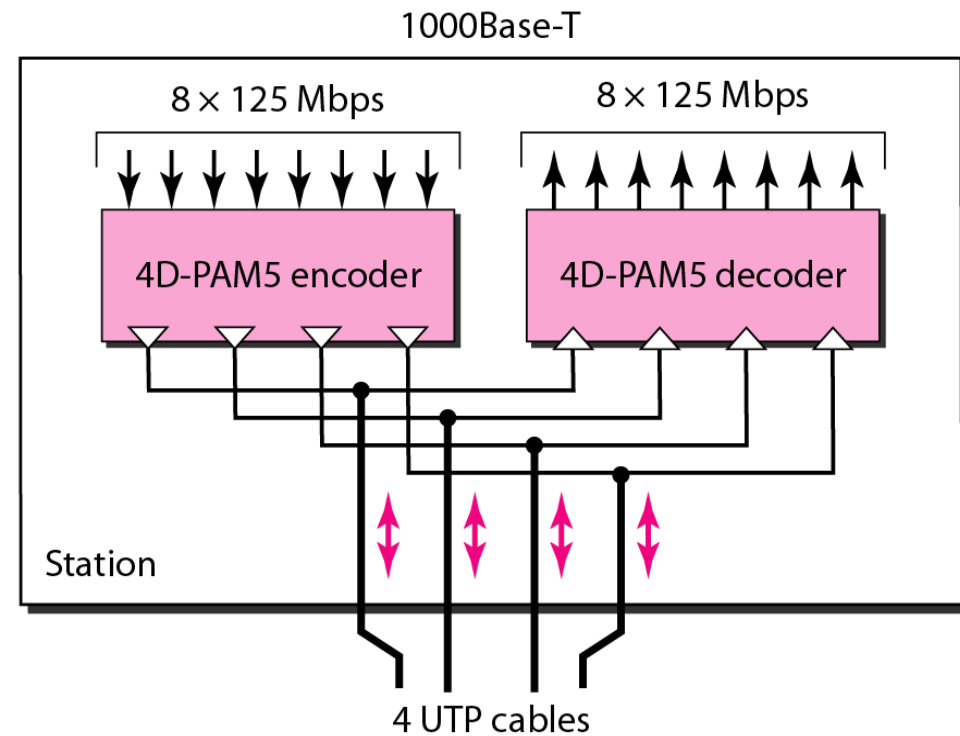
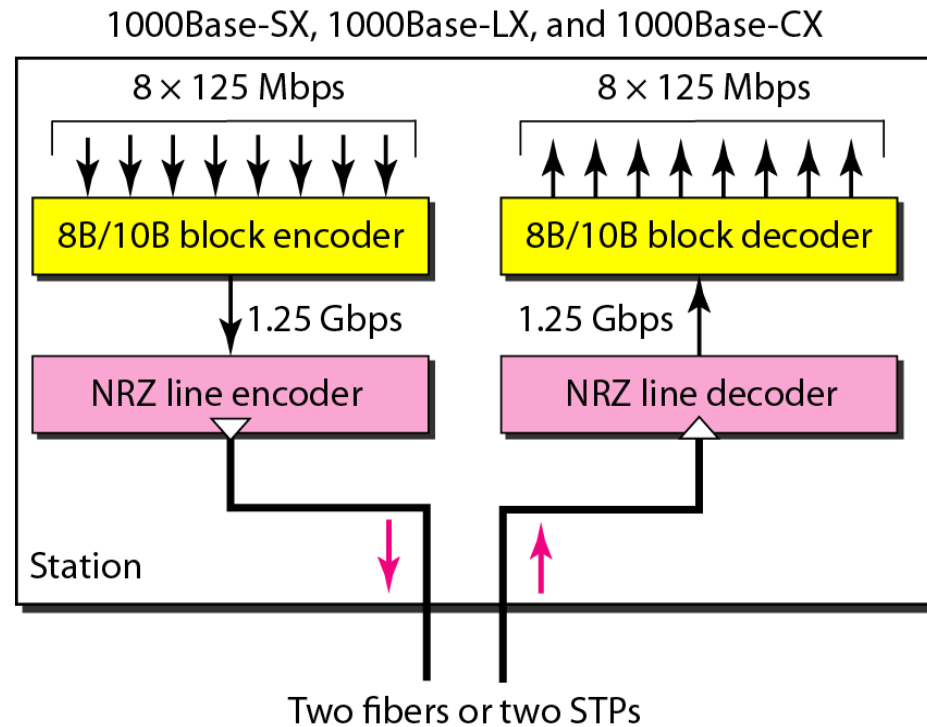
<i>Characteristics</i>	<i>100Base-TX</i>	<i>100Base-FX</i>	<i>100Base-T4</i>
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	
Line encoding	MLT-3	NRZ-I	8B/6T

Gigabit Ethernet

- The need for an even higher data rate resulted in the design of the Gigabit Ethernet protocol (1000 Mbps).
- The IEEE committee calls the standard 802.3z.



Encoding in Gigabit Ethernet Implementations



Summary of Gigabit Ethernet Implementations

<i>Characteristics</i>	<i>1000Base-SX</i>	<i>1000Base-LX</i>	<i>1000Base-CX</i>	<i>1000Base-T</i>
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5

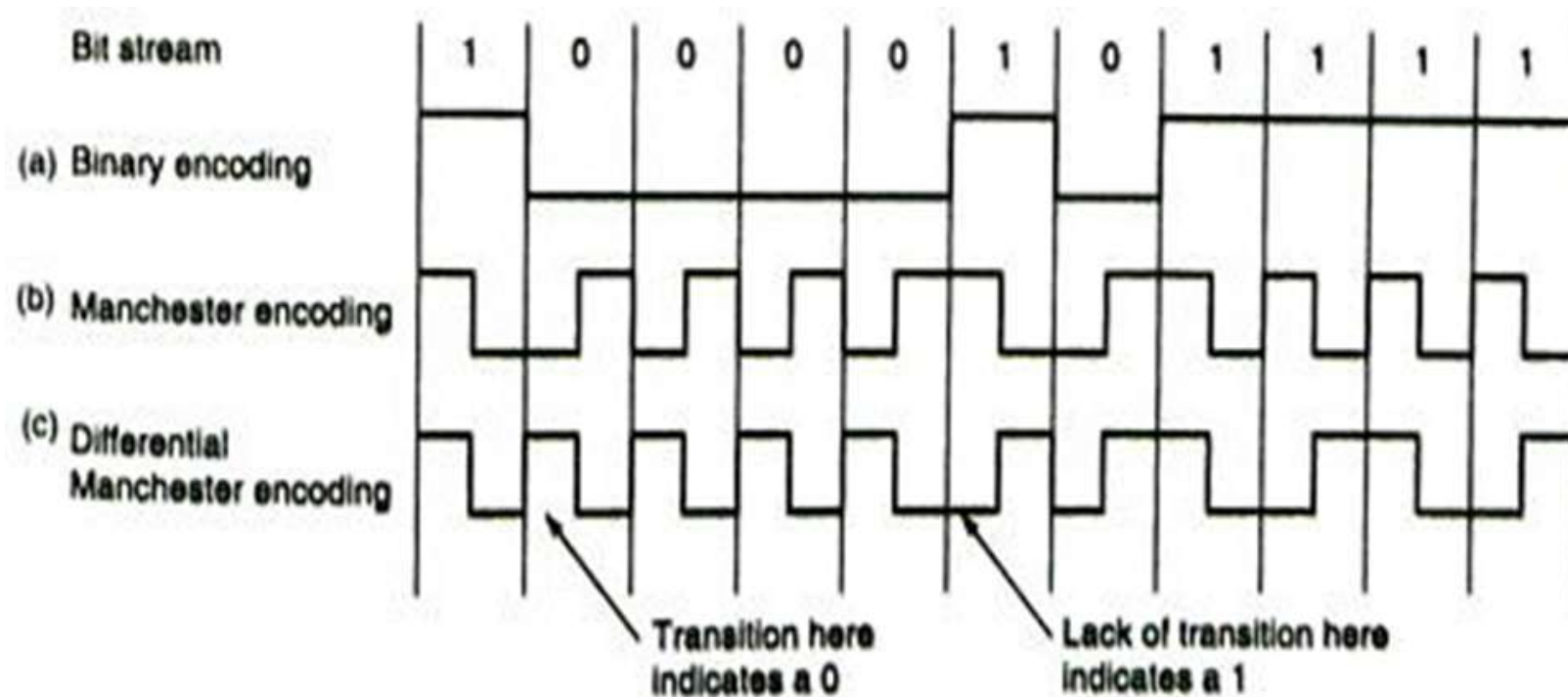
- Summary of 10-Gigabit Ethernet implementations

<i>Characteristics</i>	<i>10GBase-S</i>	<i>10GBase-L</i>	<i>10GBase-E</i>
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-nm single mode
Maximum length	300 m	10 km	40 km

- 40G & 100 G Ethernet: The IEEE 802.3ba, a new standard governing 40G & 100G Ethernet was finalized in June 2010.

Manchester Encoding

- Ethernet uses Manchester encoding to transmit data on the LAN.
- With Manchester encoding, each bit contains a transition; a 1 has a transition from up to down, whereas a 0 has a transition from down to up.



- Each Ethernet's adapter runs the CSMA/CD protocol without explicit coordination with the other adapters on the Ethernet.
 - The adapter obtains a network-layer PDU from its parent node, prepares an Ethernet frame, and put the frame in an adapter buffer.
 - The adapter senses the channel: if it is idle, the adapter starts to transmit the frame; if it is busy, the adapter waits until the channel is idle, then starts to transmit the frame.
 - While transmitting, the adapter monitors for the presence of signal energy coming from other adapters. If the adapter transmits the entire frame without collision detection, the transmission is successful.
 - If the adapter detects a collision while transmitting, it stops transmitting its frame and instead transmits a 48-bit jam signal.
 - After aborting, the adapter waits a random time (using Binary Exponential Back-off algorithm) and retry.



- When a collision occurs, the current transmission is aborted and the adapter waits a random time to retry.
- Binary Exponential Back-off Algorithm is used to choose a random time dynamically.
 - After the first collision, each station waits either 0 or 1 slot time before trying again. If each one picks the same random number, they collide again.
 - After the second collision, each one picks either 0, 1, 2, or 3 at random and waits that number of slot times.
 - In general, after i collisions, a random number between 0 and $2^i - 1$ is chosen, and the adapter waits that number of slot times. However, after ten collisions have been reached, the randomization interval is frozen at a maximum of $(2^{10} - 1)$ slots.
 - After 16 collisions, the controller gives up and reports failure back to the computer.

Token Ring (IEEE 802.5)

- Token-passing networks move a small frame, called a token, around the network
- Possession of the token grants the right to transmit.
- If a node receiving the token has no information to send, it passes the token to the next end station
- If a station possessing the token does have information to transmit
 - It seizes the token, alters 1 bit of the token (which turns the token into a start-of-frame sequence),
 - Appends the information that it wants to transmit, and sends this information to the next station on the ring.
- Each station can hold the token for a maximum period of time

Token Ring (IEEE 802.5)

- Priority System
 - Token Ring frames have two fields that control priority: the priority field and the reservation field.
 - Only stations with a priority equal to or higher than the priority value contained in a token can seize that token.
 - Only stations with a priority value higher than that of the transmitting station can reserve the token for the next pass around the network.
- Fault-Management Mechanisms
 - One station in the Token Ring network is selected to be the active monitor.
 - This station acts as a centralized source of timing information for other ring stations and performs a variety of ring-maintenance functions.
- Frame format: Support two basic frame types: tokens and data/command frames.

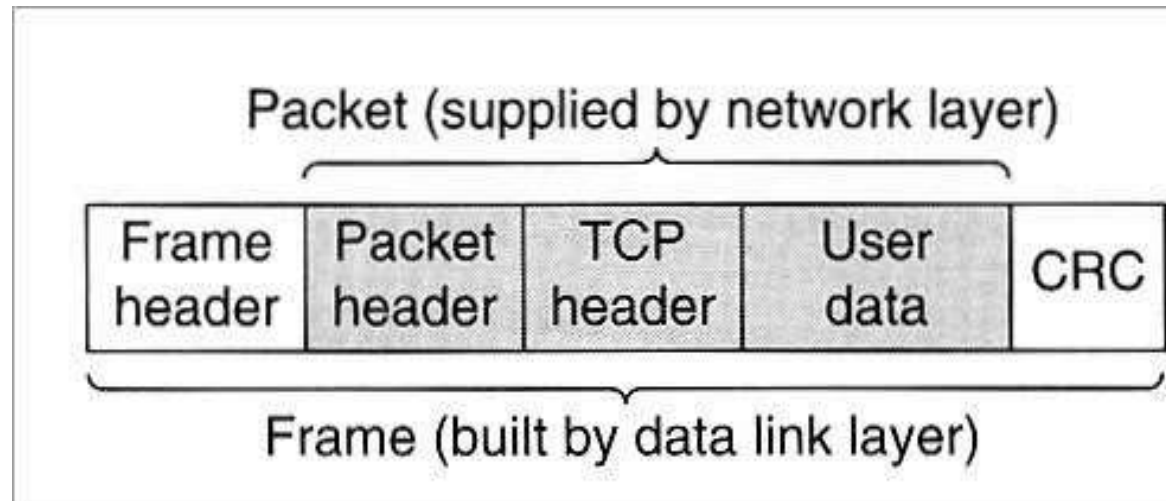
Hub, Bridge, Router, and Switch



- Hub: hub is a repeater with some additional network management functionality (such as performance or accounting management)
- Bridge: bridge operates on Ethernet frames and thus a layer-2 device. It does the following two functions:
 - Filtering: determine whether a frame should be forwarded to some interface or should just be dropped.
 - Forwarding: determine the interfaces to which a frame should be directed.

Hub, Bridge, Router, and Switch

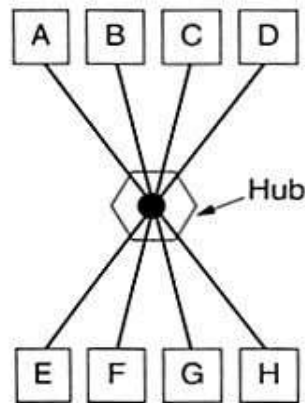
- Router: router is a store-and-forward packet switch that forward packets using network-layer addresses (layer-3).
- Although a bridge is also a store- and-forward packet switch, it forwards packets using LAN addresses.



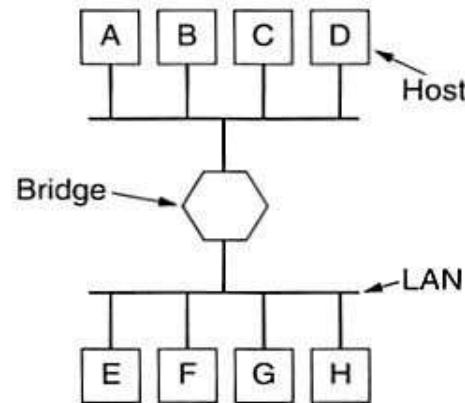
- As a network administrator, how to choose between bridge and router?

Hub, Bridge, Router, and Switch

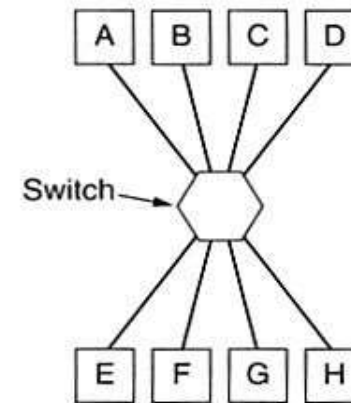
- Switch: switch is in essence a high-performance multi-interface bridge. The difference between a bridge and switch:
 - Bridges usually have two or four interfaces, whereas switches have dozens of interfaces (e.g., 24 ports).
 - Switches are usually used to connect individual computers, and operate in a full-duplex mode.



(a)



(b)



(c)

(a) A hub. (b) A bridge. (c) A switch.

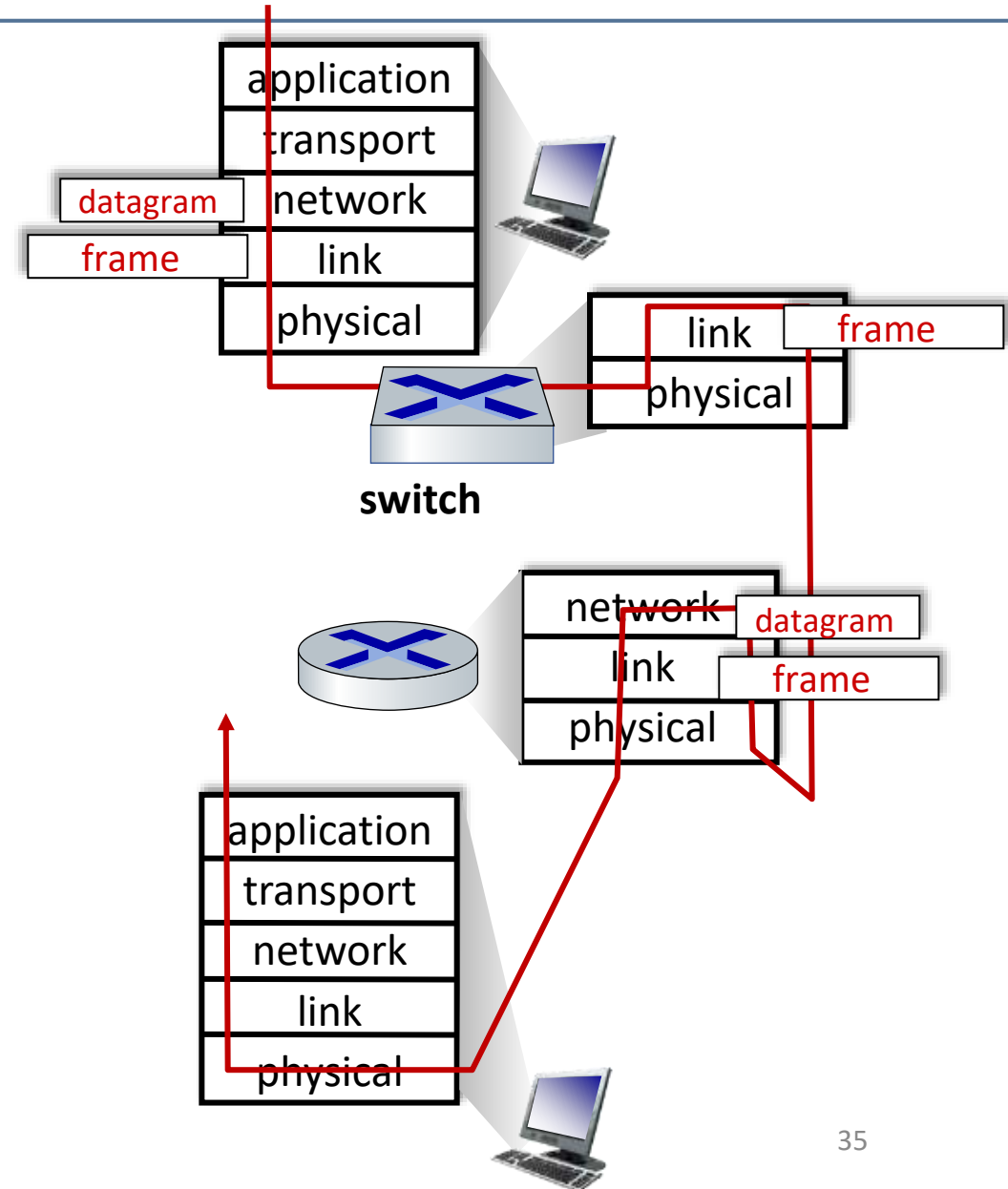
Switches vs. routers

both are store-and-forward:

- *routers*: network-layer devices (examine network-layer headers)
- *switches*: link-layer devices (examine link-layer headers)

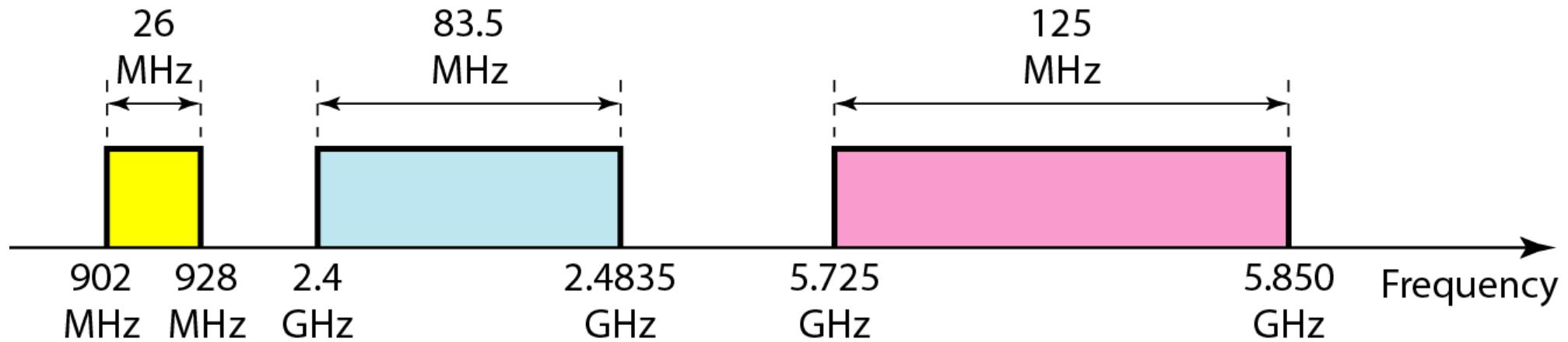
both have forwarding tables:

- *routers*: compute tables using routing algorithms, IP addresses
- *switches*: learn forwarding table using flooding, learning, MAC addresses



Wireless LANs (IEEE 802.11)

- IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data link layers.
- Industrial, scientific, and medical (ISM) band



Wireless LANs (IEEE 802.11)

- Currently, the most popular LAN standards are 802.11b (11 Mbps), 802.11g (54 Mbps) and 802.11n (600Mbps).
- Channel Allocation in Different Countries

Channel	Frequency (MHz)	North America	Japan	Most of world
1	2412	Yes	Yes	Yes
2	2417	Yes	Yes	Yes
3	2422	Yes	Yes	Yes
4	2427	Yes	Yes	Yes
5	2432	Yes	Yes	Yes
6	2437	Yes	Yes	Yes
7	2442	Yes	Yes	Yes
8	2447	Yes	Yes	Yes
9	2452	Yes	Yes	Yes
10	2457	Yes	Yes	Yes
11	2462	Yes	Yes	Yes
12	2467	No	Yes	Yes
13	2472	No	Yes	Yes
14	2484	No	.11b only	No

Wireless LANs (IEEE 802.11)

802.11 network standards v · d · e [hide]										
802.11 Protocol	Release ^[7]	Freq. (GHz)	Bandwidth (MHz)	Data rate per stream (Mbit/s) ^[8]	Allowable MIMO streams	Modulation	Approximate indoor range ^[citation needed]		Approximate Outdoor range ^[citation needed]	
							(m)	(ft)	(m)	(ft)
–	Jun 1997	2.4	20	1, 2	1	DSSS, FHSS	20	66	100	330
a	Sep 1999	5	20	6, 9, 12, 18, 24, 36, 48, 54	1	OFDM	35	115	120	390
		3.7 ^[y]					--	--	5,000	16,000 ^[y]
b	Sep 1999	2.4	20	5.5, 11	1	DSSS	38	125	140	460
g	Jun 2003	2.4	20	6, 9, 12, 18, 24, 36, 48, 54	1	OFDM, DSSS	38	125	140	460
n	Oct 2009	2.4/5	20	7.2, 14.4, 21.7, 28.9, 43.3, 57.8, 65, 72.2 ^[z]	4	OFDM	70	230	250	820 ^[9]
			40	15, 30, 45, 60, 90, 120, 135, 150 ^[z]			70	230	250	820 ^[9]

- ^y [IEEE 802.11y-2008](#) extended operation of 802.11a to the licensed 3.7 GHz band. Increased power limits allow a range up to 5000m. As of 2009, it is only being licensed in the United States by the [FCC](#).
- ^z Assumes Short [Guard interval](#) (SGI) enabled, otherwise reduce each data rate by 10%.

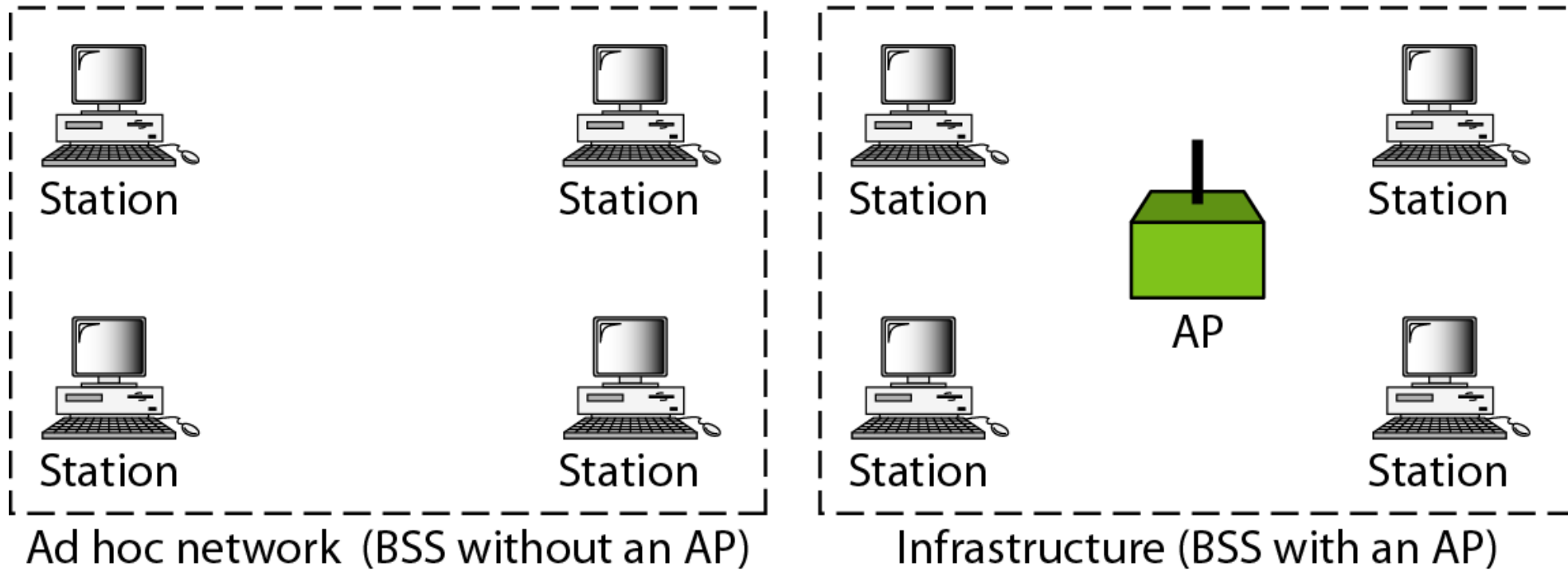
(Source: wikipedia.org)

Wireless LANs (IEEE 802.11)

- Basic service sets (BSSs)

BSS: Basic service set

AP: Access point



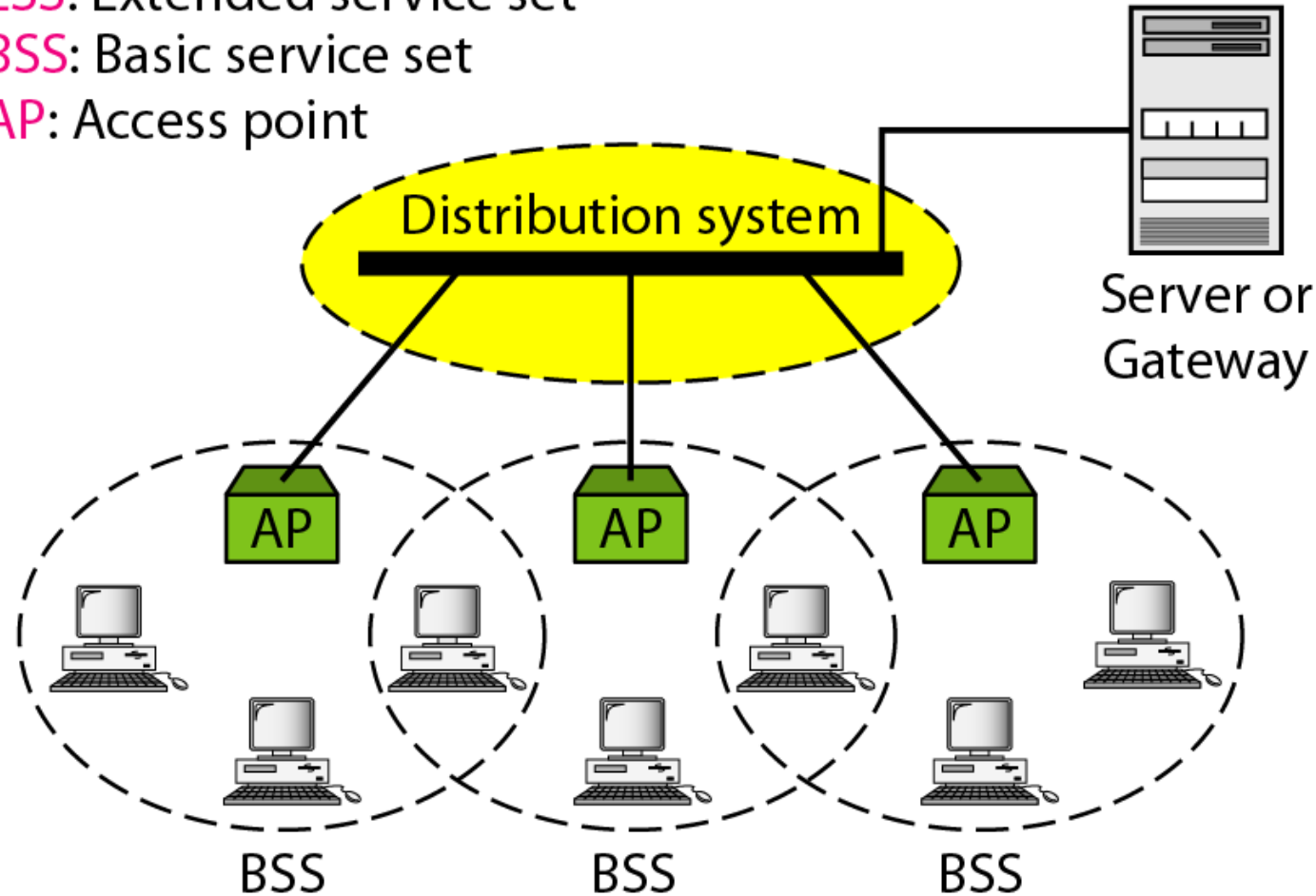
- A BSS without an AP is called an ad hoc network; a BSS with an AP is called an infrastructure network.

Extended service sets (ESSs)

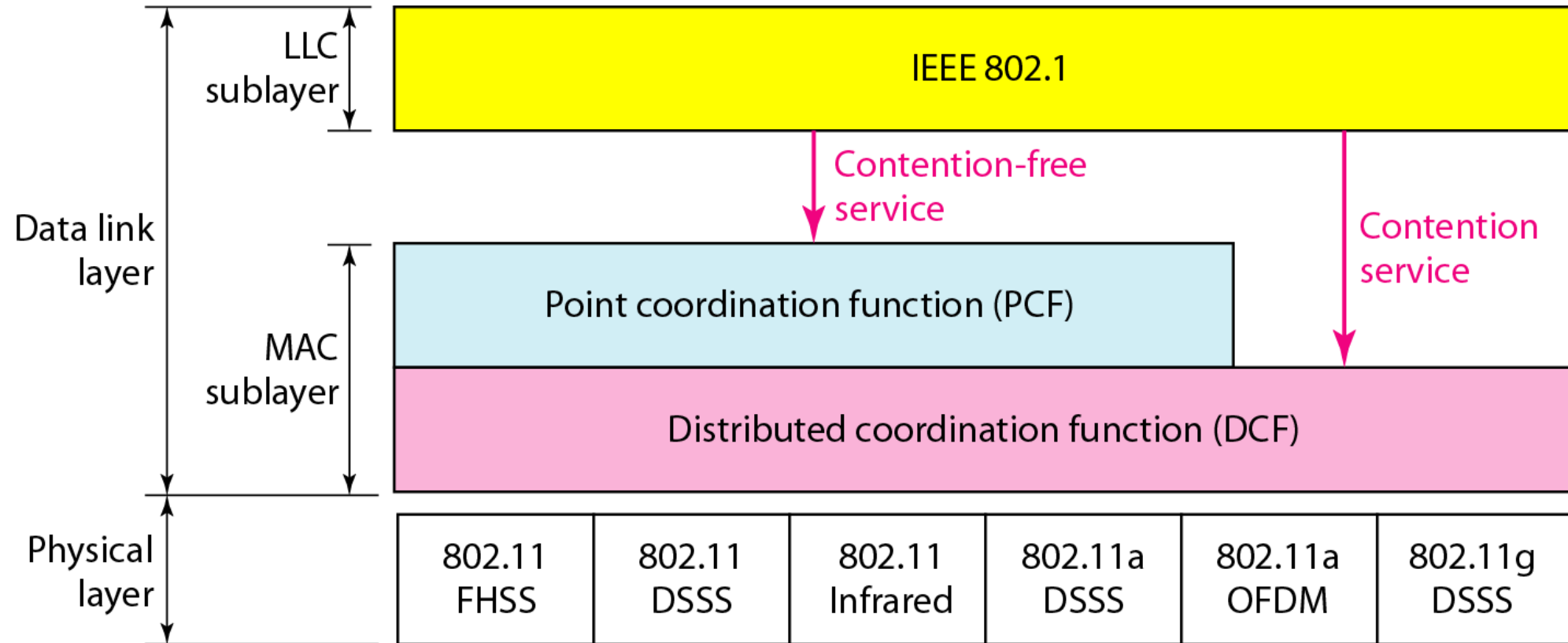
ESS: Extended service set

BSS: Basic service set

AP: Access point



MAC layers in IEEE 802.11 standard

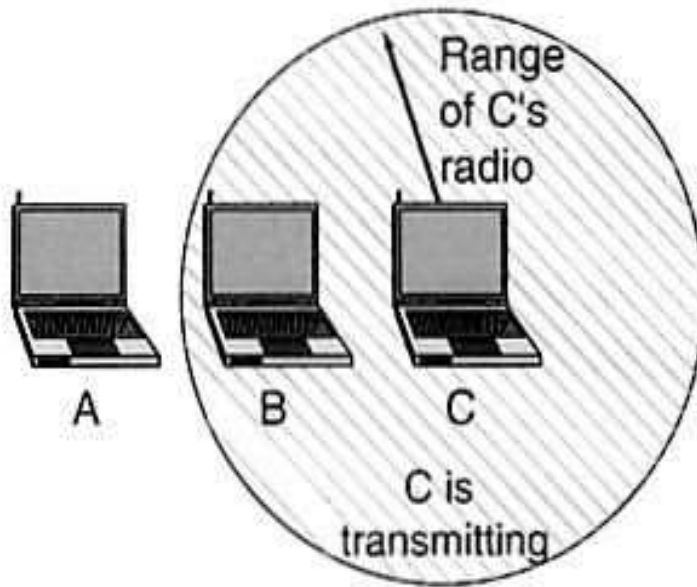


Wireless LANs (IEEE 802.11)

- MAC protocol (CSMA/CD) of Ethernet does not work in wireless LAN.
- Not all stations are within radio range of each other, transmissions going on in one part of a cell may not be received elsewhere in the same cell.
- Hidden station problem: for example, station *C* is transmitting to station *B*. If *A* senses the channel, it will not hear anything and falsely conclude that it may now start transmitting to *B*.
- Exposed station problem: for example, station *B* wants to send to *C* so it listens to the channel. When it hears a transmission, it falsely concludes that it may not send to *C*, even though *A* may be transmitting to *D*.

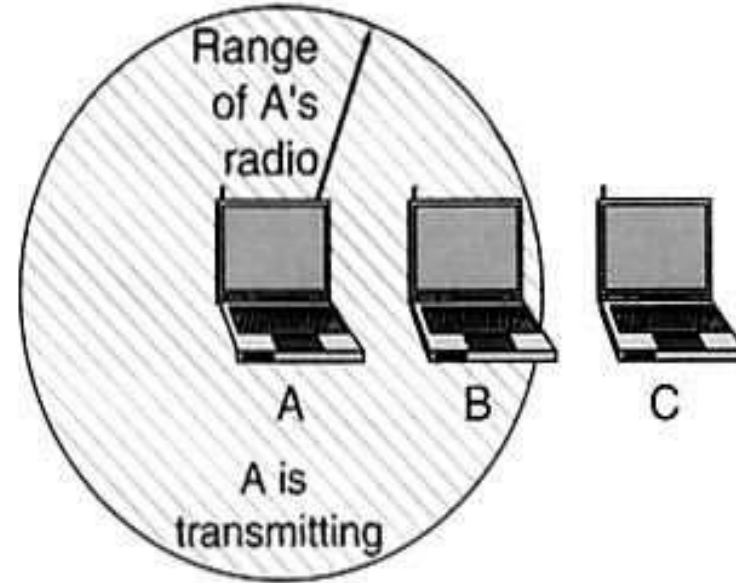
Wireless LANs (IEEE 802.11)

A wants to send to B
but cannot hear that
B is busy



(a) The hidden station problem.

B wants to send to C
but mistakenly thinks
the transmission will fail

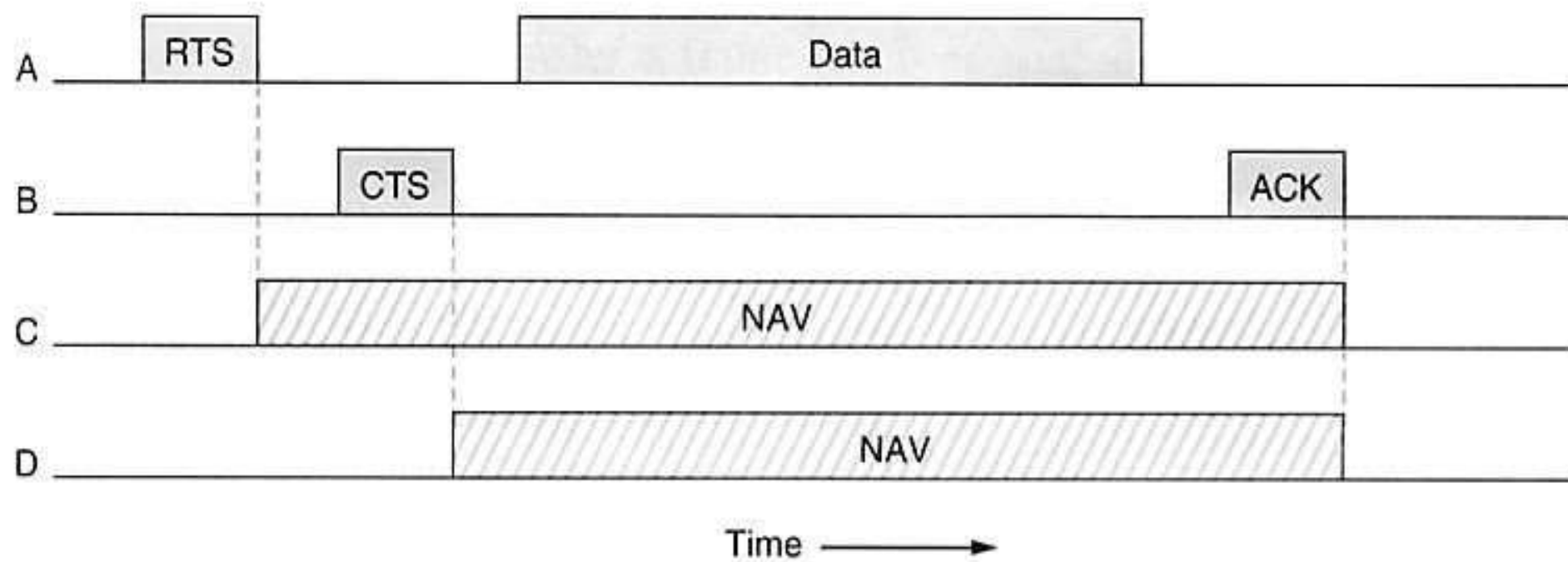


(b) The exposed station problem.

Wireless LANs (IEEE 802.11)

- 802.11 supports CSMA/CA (CSMA with collision avoidance) for MAC protocol.
 - Suppose there are 4 stations. *A* wants to send to *B*. *C* is within range of *A*. *D* is within range of *B* but not within range of *A*.
 - *A* begins by sending an RTS (request to send) frame to *B* to request permission to send.
 - If *B* decides to grant permission, it replies a CTS (clear to send) frame.
 - Upon receipt of the CTS, *A* sends its frame and starts an ACK timer.
 - Upon the correct receipt of the data frame, *B* responds with an ACK frame, terminating the exchange.
 - For *C*, it is within range of *A*, it will receive the RTS frame. Then it asserts a kind of virtual channel busy for itself, indicated by **NAV** (network allocation vector).
 - For *D*, it does not hear the RTS, but it does hear the CTS, so it also asserts the NAV signal for itself.

CSMA/CA



Wireless LANs (IEEE 802.11)

- PCF (point coordination function): the base station polls the other stations, asking them if they have any frames to send. → No collision.
- Mechanism: the base station broadcast a **beacon frame** periodically.
 - Ask other stations if they have any frame to send.
 - Invite new stations to sign up for polling service.

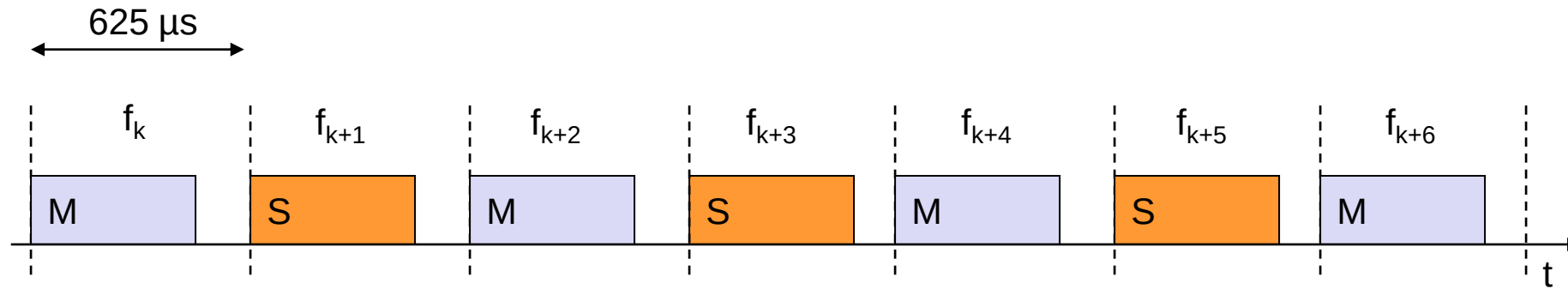
Bluetooth (IEEE 802.15)

- Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on.
- A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.
- It operates in the unlicensed 2.4 GHz spectrum.
- The basic unit of a Bluetooth system is a piconet, which consists of a master node and up to seven active slave nodes (and up to 255 parked nodes) within a distance of 10 meters normally (could be up to 100 meters).

Bluetooth (IEEE 802.15)

- Multiple piconets can exist in the same room and can be connected via a bridge node.
- The slaves just do whatever the master tells them to do.
- The data-rate ranges from 1Mbit/sec to 24Mbit/sec (depending on the versions).
- Lower power, lower cost & data encryption.

Frequency hopping



M: Master node

S: Slave node

1. In a BLE network, the master node designs a unique frequency hopping pattern ($f_k, f_{k+1}, \dots, f_{k+6}$) and notifies the slave nodes within the network.
2. Master and slave nodes are then synchronized in clock.
3. Master and slave nodes hop together based on the agreed frequency hopping pattern.